OS LAB 03

Name: Gulam Mujtaba

Roll No 24K-0535

Task 1

ali-rooman@ali-rooman-VirtualBox: ~/Desktop/24K-0535

```
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ gedit m1.sh
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ chmod +x m1.sh
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ ./m1.sh
Enter first number:
5
Enter second number:
5
Result: 25
ali-rooman@ali-rooman
```

Open — m1.sh — ~/Desktop/24K-0535

```
echo "Enter first number:"
read num1

echo "Enter second number:"
read num2

result=$((num1 * num2))

echo "Result: $result"
```

Task 2

ali-rooman@ali-rooman-VirtualBox: ~/Desktop/24K-0535

```
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ gedit m1.sh
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ chmod +x m1.sh
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ ./m1.sh
Enter first number:
5
Enter second number:
5
Result: 25
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ gedit v1.sh
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ chmod +x v1.sh
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ ./v1.sh
Enter a sentence:
Mujtaba Qureshi
Number of vowels: 6
ali-rooman@ali-rooman
```

Open — v1.sh — ~/Desktop/24K-0535

m1.sh | v1.sh

```
echo "Enter a sentence:"
read sentence

count=$(echo "$sentence" | grep -o -i "[aeiou]" | wc -l)
echo "Number of vowels: $count"
```

Task 3

```
ali-rooman@ali-rooman-VirtualBox: ~/Desktop/24K-0535
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ gedit a1.sh
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ chmod +x a1.sh
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ ./a1.sh
Enter filename:
m1.sh
Date and time appended successfully.
ali-rooman@ali-rooman-VirtualBox:~/De
```

Open a1.sh ~/Desktop/24K-0535

m1.sh | v1.sh

```
echo "Enter filename:"
read filename
if [ -f "$filename" ]
then
    echo "Date and Time: $(date)" >> "$filename"
    echo "Date and time appended successfully."
else
    echo "File does not exist."
fi
```

Task 4

```
ali-rooman@ali-rooman-VirtualBox: ~/Desktop/24K-0535
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ gedit f1.sh
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ chmod +x f1.sh
ali-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ ./f1.sh
Enter file or directory name:
m1.sh
It is a regular file.
ali-rooman@ali-rooman-VirtualBox
```

Open f1.sh ~/Desktop/24K-0535

m1.sh | v1.sh | a1.sh | file_status.sh | f1.sh

```
echo "Enter file or directory name:"
read name

if [ -e "$name" ]
then
    if [ -f "$name" ]
    then
        echo "It is a regular file."
    elif [ -d "$name" ]
    then
        echo "It is a directory."
    else
        echo "It exists but is neither a regular file nor a
directory."
    fi
else
    echo "File or directory does not exist."
fi
```

Task 5

ali-rooman@ali-rooman-VirtualBox: ~/Desktop/24K-0535

```
li-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ gedit b1.sh
li-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ chmod +x b1.sh
li-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ ./b1.sh
sage: ./b1.sh <directory_path>
li-rooman@ali-rooman-VirtualBo
```

Open — b1.sh — ~/Desktop/24K-0535

m1.sh | v1.sh | a1.sh | f1.sh | b1.sh

```
if [ $# -ne 1 ]
then
    echo "Usage: $0 <directory_path>"
    exit 1
fi
source_dir=$1
backup_base="$HOME/backup"
date=$(date +%Y-%m-%d)
backup_dir="$backup_base/backup_$date"
if [ ! -d "$source_dir" ]
then
    echo "Directory does not exist."
    exit 1
fi
```

ali-rooman@ali-rooman-VirtualBox: ~/Desktop/24K-0535

```
li-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ gedit b1.sh
li-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ chmod +x b1.sh
li-rooman@ali-rooman-VirtualBox:~/Desktop/24K-0535$ ./b1.sh
sage: ./b1.sh <directory_path>
li-rooman@ali-rooman-VirtualBo
```

Open — b1.sh — ~/Desktop/24K-0535

m1.sh | v1.sh | a1.sh | f1.sh | b1.

```
then
    echo "Directory does not exist."
    exit 1
fi
mkdir -p "$backup_dir"
cp -r "$source_dir" "$backup_dir"
file_count=$(find "$source_dir" -type f | wc -l)
dir_count=$(find "$source_dir" -type d | wc -l)S
echo "Backup completed successfully!"
echo "Source Directory: $source_dir"
echo "Backup Location: $backup_dir"
echo "Total Files Backed Up: $file_count"
echo "Total Directories Backed Up: $dir_count"
```